# Hands-On Lab: Setting up AWS KMS Key for S3 Encryption

## Objective

In this lab, you will create a customer-managed AWS KMS key and configure an Amazon S3 bucket to use this key for server-side encryption (SSE-KMS).

## Prerequisites

- AWS Account with administrator or KMS/S3 permissions
- Access to AWS Management Console
- Basic understanding of AWS IAM, S3, and KMS

## Lab Architecture Overview

Amazon S3 bucket encrypted using a customer-managed AWS KMS symmetric key.

## Step 1: Create a Customer Managed KMS Key

1. Sign in to the AWS Management Console.
2. Navigate to Services → Security, Identity & Compliance → KMS.
3. Click Create key.
4. Select Symmetric as key type and Encrypt and decrypt for key usage.
5. Click Next.
6. Enter Alias (example: alias/s3-encryption-key).
7. (Optional) Add a description and tags.
8. Click Next.
9. Select key administrators (e.g., your IAM user or role).
10. Click Next.
11. Select key users (IAM roles/users or the S3 service if required).
12. Click Finish.

## Step 2: Create an S3 Bucket

1. Navigate to Services → Storage → S3.
2. Click Create bucket.
3. Enter a globally unique bucket name.
4. Choose an AWS region (same region as KMS key).
5. Leave other settings default or as per organizational policy.
6. Click Create bucket.

## Step 3: Enable Default Encryption Using KMS

1. Open the created S3 bucket.
2. Go to Properties tab.
3. Scroll to Default encryption and click Edit.
4. Choose Enable.
5. Select Server-side encryption with AWS KMS keys (SSE-KMS).

6. Choose Select from your AWS KMS keys.
7. Select the key alias created earlier.
8. Click Save changes.

## Step 4: Upload and Validate Encryption

1. Upload a file to the S3 bucket.
2. Select the uploaded object.
3. Go to the Properties tab.
4. Verify the Server-side encryption shows AWS-KMS.

## Step 5: Optional CLI Validation

aws s3api head-object --bucket <bucket-name> --key <object-name>
Verify SSEKMSKeyId in the response.

## Cleanup

1. Delete objects in the S3 bucket.
2. Delete the S3 bucket.
3. Schedule deletion of the KMS key.

## Outcome

You have successfully configured S3 default encryption using a customer-managed AWS KMS key.